Book

Lies, damn lies, and scientific research

As an editor of this medical journal, I try to judge whether the research reported in a paper was well done and whether the paper reports that research accurately. In general, scientists believe in their work, and, try as they might to maintain equipoise, find it hard to keep their biases—of which they might well be unaware—from putting a spin on their results. It's only human. In these cases, most researchers will tone down claims when pressed by peer reviewers and editors. But others are so wedded to their hypothesis that their papers become polemics. Negotiations with these authors can be difficult, but at least the disagreements are open and the discussions civil, usually.

But, as Horace Freeland Judson describes in *The Great Betrayal: Fraud in Science* there are researchers who submit papers in which data have been faked, numbers fudged, and ideas filched. How many such papers come in? How many slip through the peer review and editing process? It's impossible to know. In the past 4 years, *The Lancet* has had one proven case of fraud. Judson, a journalist and historian, would argue that it is likely we've missed many more

Judson thinks that fraud is common because it is the inevitable product of the current culture of science. Fraud, he says, is intrinsic in institutional cultures that are "characterized by secrecy, privilege, lack of accountability". Scientists have won exactly such a privileged place in society by presenting themselves as altruistic seekers of truth whose method will invariably ferret out not only truth, but also fraud. "The grandees of the scientific establishment", writes Judson, "regularly proclaim that scientific fraud is vanishingly rare and that perpetrators are isolated individuals who act out of a twisted psychopathology".

The elevation of the scientific profession to the status it holds today in the USA can be credited at least in part to Vannear Bush, an electrical engineer, who served in President Franklin Roosevelt's administration as director of the Office of Research and Development. This agency was established in World War II to adapt the discoveries of basic scientists and the tools of civilian technology to the war effort. Basic research had already led to improved radar systems, mass production of sulfa and penicillin antibiotics, and, of course, the atomic bomb. Even greater advances would come, Bush argued in a 1945 proposal to Roosevelt's successor President Harry Truman, if the government funded scientists and gave them free rein to pursue questions of their own choosing. "Scientific progress on a broad front", Bush wrote, "results from the free play of free intellects, working on subjects of their own choice, in the manner dictated by their curiosity, of the unknown".

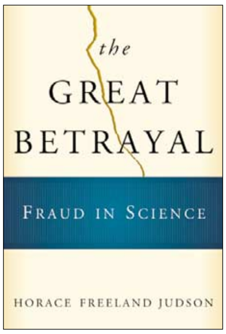
"Judson paints a dark picture of science today, but we may see far darker days ahead as proof and profit become inextricably mixed"

Since then, US investment in research has climbed steadily, and in recent years, steeply: in 1995, for example, the US National Institutes of Health (NIH) budget was US$67·1 million; this year, under a decade later, the NIH budget is nearly $28 billion. To protect this investment, Judson charges, scientists play down the extent and seriousness of fraud. "In sum", he says, "the scientific communities believe that public funding is their right, but so is freedom from public control. The grandees, the guardians, must defend the honor of science: fraud, if it proved to be more than rare and individual, perhaps even to be built into the ways institutions of science work, would engender distrust among legislators and the public and so threaten the money stream and self-governance."

Fraud has always been present in science, long before the NIH was handing out grants. Gregor Mendel, the founder of genetics, had results that are just too good to be true. Louis Pasteur's notebooks, long kept secret, reveal that he misled the world and his fellow scientists about the research behind two of his most famous experiments: the vaccination of sheep against anthrax, and that of a boy against rabies. And it is common knowledge that Sigmund Freud fabricated many of the case studies on which he built his psychoanalytic theories and career.

Judson recounts, in detail, a number of well publicised cases of scientific fraud that have come to light over the past two decades. "In almost every case, to be sure, some one individual gets blamed", he writes, "but these frauds cannot be presented even as anecdotes without an accounting of the relationships among many people within the laboratory and the larger institutional setting. The cases exhibit multiple, tangled complicities."

These complicities tend to fall into predictable syndromes, Judson maintains. "The dominant form is *the prodigy*; others are the *mentor seduced*, the *folie à deux*, and the *arrogance of power*." The prodigy, for example, is a young researcher whose productivity is too good to be true. One such prodigy was John Darsee, a Harvard researcher who had, by the age of 33, published more than 125 research articles, book chapters, abstracts, and other papers. But it turned out that he had fabricated data in scores of papers.



The Great Betrayal: Fraud in Science

Horace Freeland Judson. Harcourt, 2004. Pp 463. $28·00. ISBN 0-15-100877-9.

Some of the problems with his data should have been obvious at the time. In one paper he described a family in which a 17-year-old male had four children—the eldest of which was 8 years old. Do the maths. These and other glaring problems eluded not only the peer reviewers and editors, but also, apparently, his coauthors.

What is particularly worrying about Darsee's case and many of the others Judson describes is that the fraud was not detected by safeguards built into the scientific system, such as peer review. Instead, deception is typically discovered by dumb luck: a co-worker notices something strange in a log book; a colleague stumbles across a discrepancy in a table. Closer inspection often reveals fraud so ham-handed that the perpetrator seems to have been asking to be caught.

So what can be done? Attempts, some more promising than others, have been made to combat fraud. Journals, for example, have tightened their rules, requiring all authors to have made significant contributions to a paper that carries their name, and to vouch for its content. Institutions have established much-needed formal procedures to expedite inquiries into fraud allegations. In the past, some institutions seem to have been more concerned in hushing up scandal than in rooting out fraud; whistle-blowers have found their careers destroyed while department heads have escaped rebuke. Furthermore, Judson thinks that e-publishing, which makes it possible to post not only the paper on the internet, but also the original manuscript, peer reviewers' comments, and raw data will have a substantial effect on fraud. But while e-publishing will make plagiarism and duplicate publication easy to detect, whether it will prevent the determined fabricator is not so certain.

Most worrisome of all, most of the fraud Judson describes was done by individuals seeking acceptance and advancement in the world of science. Increasingly, a great deal of research is in fact product development; researchers often have more than their egos and grants on the line—a fortune can rest on their results. Judson paints a dark picture of science today, but we may see far darker days ahead as proof and profit become inextricably mixed.

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In brief

Future Face

A Wellcome Trust exhibition at the Science Museum, Exhibition Road, London, UK. Until Feb 13, 2005. Free.

Exhibition Same old faces

What, asks this exhibition, might our faces look like in the future? From what's on display, I'm betting that our faces will look much as they do now, our imperfect physiognomies being vastly preferable to the vacuous, virtual reality of computerised alternatives.

For example, the limited facial expression of a digital Marlene Dietrich comes nowhere near to matching the unique reality of the woman. More interesting than such botched, digital duplicity is the creativity evident in video clips of children trying to express themselves using only facial gestures. And infinitely more moving is archive material documenting Harold Gillies' work with artists and his surgical team's use of their sketches in the reconstruction of ruined faces of servicemen in World War I.

Although the curator has ransacked storerooms to display some very interesting material, the thematic spin is a flawed attempt to court controversy. It seems self-evident that the individuality of the human countenance will outlast the electronic age. *Future Face* leaves you, as a contemporary advertisement exhorts, happy to "Love the skin you're in", rather than pining for some digital concept of perfection.

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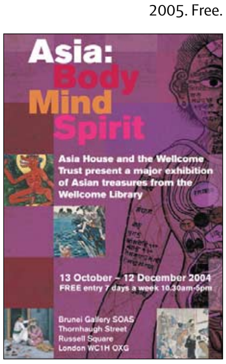

Asia: Body Mind Spirit

A Wellcome Trust and Asia House exhibition at the Brunei Gallery, School of Oriental and African Studies (SOAS), Thornhaugh Street, London, UK. Until Dec 12, 2004. Free.

Exhibition The whole story

Just before hopping on a bus to the Brunei gallery, I grabbed a copy of *The Guardian* newspaper. An article on eating out caught my eye: in east London, an ayurvedic restaurant is serving bespoke menus based on in-house analysis of customers' health.

Ayurveda forms part of an exhibition that is holistic in its subject matter and incorporation of both artistic and medical perspectives. Asian medical traditions, at least those in India, China, Japan, and Tibet, share many features. Blocking a "life force" (qi in China) causes illness and harmony is crucial to good health—treatments are designed to balance "humours", such as phlegm and bile.

The curators have created a space that feels as much art gallery as history of medicine exhibit, but substance isn't sacrificed for beauty. The interchange of theories between East and West is outlined and practices such as acupuncture and feng shui contextualised. A screen flickering with images of a woman practising tai chi urges visitors to join in. Inhibited by the presence of a bored security guard, I moved on. A mirror stands by a chart describing how to assess your internal organs. My self-diagnosis was sobering. Kidneys: fine. Heart: not bad. Liver: has seen better days. Perhaps it's time to book a table at the ayurvedic restaurant.

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